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Queue analysis apparatus, control method, and non-transitory computer-readable medium

Abstract

A queue analysis apparatus (2000) estimates a position and an orientation of each object (20) included in a target image (10). The target image (10) is generated by a camera (50) that captures the object (20). The queue analysis apparatus (2000) generates a queue line (40) that expresses, by a linear shape, a queue included in a queue region (30) being a region representing a queue in the target image (10), based on a position and an orientation being estimated for each object (20) included in the queue region (30).

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Background/Summary

(1) This application is a National Stage Entry of PCT/JP2020/004934 filed on Feb. 7, 2020, the contents of all of which are incorporated herein by reference, in their entirety.

TECHNICAL FIELD

(2) The present invention relates to an analysis of a queue of objects.

BACKGROUND ART

(3) A technique for acquiring information related to a queue of people and the like from an image

has been developed. For example, Patent Document 1 discloses a technique for detecting a pixel representing a stationary object from a time-series image, forming the pixels into a group by using advance knowledge related to a shape of a queue, and thus detecting a queue from an image.

RELATED DOCUMENT

Patent Document

(4) [Patent Document 1] Japanese Patent Application Publication No. 2016-224651

DISCLOSURE OF THE INVENTION

Technical Problem

(5) An image processing apparatus in Patent Document 1 requires advance knowledge related to a shape of a queue, such as an extending direction (a direction in which a queue extends) of a queue. Thus, a user of the image processing apparatus in Patent Document 1 requires labor for recognizing a shape of a queue in advance, and providing information related to the recognized shape to the image processing apparatus.

(6) The present invention has been made in view of the problem described above, and one of objects of the present invention is to provide a technique capable of recognizing a state of a queue from an image with less labor.

Solution to Problem

(7) A queue analysis apparatus according to the present invention includes: 1) an estimation unit that estimates a position and an orientation of an object included in a target image; and 2) a queue line generation unit that generates, for a queue region being a region including a queue of an object in the target image, a queue line representing a queue included in the queue region by a line, based on the position and the orientation being estimated for the object included in the queue region.

(8) A control method according to the present invention is executed by a computer. The control method includes: 1) an estimation step of estimating a position and an orientation of an object included in a target image; and 2) a queue line generation step of generating, for a queue region being a region including a queue of an object in the target image, a queue line representing a queue included in the queue region by a line, based on the position and the orientation being estimated for the object included in the queue region.

(9) A program according to the present invention causes a computer to execute the control method according to the present invention.

Advantageous Effects of Invention

(10) According to the present invention, a technique capable of recognizing a state of a queue from an image with less labor is provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram illustrating an outline of an operation of a queue analysis apparatus according to the present example embodiment.

(2) FIG. 2 is a diagram illustrating a configuration of the queue analysis apparatus according to an example embodiment 1.

(3) FIG. 3 is a diagram illustrating a computer for achieving the queue analysis apparatus.

(4) FIG. 4 is a flowchart illustrating a flow of processing performed by the queue analysis apparatus according to the example embodiment 1.

(5) FIG. 5 is a diagram illustrating a direction vector computed for each object.

(6) FIG. 6 is a diagram illustrating order information in a table format.

(7) FIG. 7 is a diagram illustrating a queue line generated by connecting direction vectors.

(8) FIG. 8 is a diagram illustrating a queue line generated by connecting middle points of the direction vectors.

- (9) FIGS. **9A** and **9B** are diagrams illustrating a queue line acquired by linear approximation.
- (10) FIGS. **10A** and **10B** are diagrams illustrating a case where non-linear interpolation processing is performed in generation of a queue line.
- (11) FIGS. **11A** to **11E** are diagrams illustrating a shape that can be used as a queue line.
- (12) FIG. **12** is a block diagram illustrating a functional configuration of the queue analysis apparatus including an output information generation unit.
- (13) FIG. **13** is a diagram illustrating a result image.
- (14) FIG. **14** is a diagram illustrating a result image indicating a front position and an end position of a queue.
- (15) FIG. **15** is a diagram illustrating a result image indicating a waiting time at an end.
- (16) FIG. **16** is a diagram illustrating a result image also indicating a waiting time in a position other than an end.
- (17) FIG. **17** is a diagram illustrating a result image including a past queue line.
- (18) FIG. **18** is a diagram illustrating a method of estimating a future queue line.

DESCRIPTION OF EMBODIMENTS

(19) Hereinafter, example embodiments of the present invention will be described with reference to the drawings. Note that, in all of the drawings, a similar component has a similar reference sign, and description thereof will not be repeated as appropriate. Further, in each block diagram, each block represents a configuration of a functional unit instead of a configuration of a hardware unit unless otherwise described.

(20) <Outline>

(21) FIG. **1** is a diagram illustrating an outline of an operation of a queue analysis apparatus **2000** according to the present example embodiment. FIG. **1** is a schematic diagram representing description for facilitating understanding of the operation of the queue analysis apparatus **2000**, and does not specifically limit the operation of the queue analysis apparatus **2000**.

(22) The queue analysis apparatus **2000** generates a queue line **40** that expresses a queue included in a target image **10** by a linear shape. To do so, the queue analysis apparatus **2000** estimates a position and an orientation of each object **20** included in the target image **10**. The target image **10** is generated by a camera **50** that captures the object **20**. Note that, an example in which capturing is performed from immediately above is exemplified for the target image **10** illustrated in FIG. **1**, but a capturing angle of view of the target image **10** is not limited, and capturing may be performed from any angle of view.

(23) As the object **20**, any object whose position may move can be handled. For example, as the object **20**, a person and other animals, and a car, a motorcycle, and other vehicles can be handled. Herein, a group of a plurality of objects instead of one object may be handled as the object **20**. This is, for example, a case where a mass of a few people is handled as one object **20**, or the like. Conversely, a part of one object may be handled as the object **20**. This is, for example, a case where only a head of a person instead of a whole of a person is handled as the object **20**, or the like.

(24) The queue analysis apparatus **2000** generates the queue line **40** that expresses, by a linear shape, a queue included in a queue region **30** being a region representing a queue in the target image **10**, based on a position and an orientation being estimated for each object **20** included in the queue region **30**. Herein, processing of determining the queue region **30** from the target image **10** may be performed by the queue analysis apparatus **2000**, or may be performed by an apparatus other than the queue analysis apparatus **2000**. Hereinafter, in order to simplify description, the processing of determining the queue region **30** is assumed to be performed by the queue analysis apparatus **2000** unless otherwise specified.

(25) <Representative Advantageous Effect>

(26) The queue analysis apparatus **2000** estimates a position and an orientation of each object **20** included in the target image **10**, and generates the queue line **40** representing a shape of a queue of the objects **20**, based on the estimated position and the estimated orientation of the object **20**.

According to this configuration, advance knowledge related to a shape of a queue, such as a direction in which a queue extends, may not be provided to the queue analysis apparatus **2000**. Thus, a user of the queue analysis apparatus **2000** can recognize a state of a queue from the target image **10** with less labor.

(27) Further, since advance knowledge related to a shape of a queue may not be provided, the queue line **40** can be generated from the target image **10** even in a case where a shape of a queue is not known in advance and when a queue has an unexpected shape.

(28) Hereinafter, the queue analysis apparatus **2000** according to the present example embodiment will be described in more detail.

(29) <Example of Functional Configuration of Queue Analysis Apparatus **2000**>

(30) FIG. **2** is a diagram illustrating a configuration of the queue analysis apparatus **2000** according to an example embodiment 1. The queue analysis apparatus **2000** includes an estimation unit **2020** and a queue line generation unit **2040**. The estimation unit **2020** estimates a position and an orientation of the object **20** included in the target image **10**. The queue line generation unit **2040** generates, for the queue region **30** in the target image **10**, the queue line **40** that expresses, by a line, a queue included in the queue region **30**, based on a position and an orientation being estimated for each object **20** included in the queue region **30**.

(31) <Hardware Configuration of Queue Analysis Apparatus **2000**>

(32) Each functional component unit of the queue analysis apparatus **2000** may be achieved by hardware (for example, a hard-wired electronic circuit, and the like) that achieves each functional component unit, and may be achieved by a combination of hardware and software (for example, a combination of an electronic circuit and a program that controls the electronic circuit, and the like). Hereinafter, a case where each functional component unit of the queue analysis apparatus **2000** is achieved by the combination of hardware and software will be further described.

(33) For example, the queue analysis apparatus **2000** is achieved by one computer. FIG. **3** is a diagram illustrating a computer **1000** for achieving the queue analysis apparatus **2000**. The computer **1000** is any computer. For example, the computer **1000** is a personal computer (PC), a server machine, a tablet terminal, a smartphone, or the like. The computer **1000** may be a dedicated computer designed for achieving the queue analysis apparatus **2000**, and may be a general-purpose computer.

(34) The computer **1000** includes a bus **1020**, a processor **1040**, a memory **1060**, a storage device **1080**, an input/output interface **1100**, and a network interface **1120**. The bus **1020** is a data transmission path for allowing the processor **1040**, the memory **1060**, the storage device **1080**, the input/output interface **1100**, and the network interface **1120** to transmit and receive data with one another. However, a method of connecting the processor **1040** and the like to each other is not limited to bus connection. The processor **1040** is a processor such as a central processing unit (CPU), a graphic processing unit (GPU), or a field-programmable gate array (FPGA). The memory **1060** is a main storage apparatus achieved by using a random access memory (RAM) and the like. The storage device **1080** is an auxiliary storage apparatus achieved by using a hard disk drive, a solid state drive (SSD), a memory card, a read only memory (ROM), or the like. However, the storage device **1080** may be constituted by hardware similar to hardware constituting the main storage apparatus, such as the RAM.

(35) The input/output interface **1100** is an interface for connecting the computer **1000** and an input/output device. The network interface **1120** is an interface for connecting the computer **1000** to a communication network. The communication network is, for example, a local area network (LAN) and a wide area network (WAN). A method of connection to the communication network by the network interface **1120** may be wireless connection or wired connection.

(36) The storage device **1080** stores a program module that achieves each functional component unit of the queue analysis apparatus **2000**. The processor **1040** achieves a function associated with each program module by reading each of the program modules to the memory **1060** and executing

the program module.

(37) The queue analysis apparatus **2000** may be achieved by two or more computers. Each of the computers in this case also has the hardware configuration illustrated in FIG. 3, for example.

(38) Herein, at least a part of a function of the queue analysis apparatus **2000** may be achieved by the camera **50**. In other words, the camera **50** may be used as one of computers that achieve the queue analysis apparatus **2000**. As the camera **50** that achieves at least a part of a function of the queue analysis apparatus **2000** in such a manner, for example, a camera referred to as an intelligent camera, a network camera, an Internet protocol (IP) camera, or the like can be used.

(39) <Flow of Processing>

(40) FIG. 4 is a flowchart illustrating a flow of processing performed by the queue analysis apparatus **2000** according to the example embodiment 1. The estimation unit **2020** acquires the target image **10** (S102). The estimation unit **2020** estimates the queue region **30** from the target image **10** (S104). The estimation unit **2020** estimates, for each object **20** included in the queue region **30**, a position and an orientation of the object **20** (S106). The queue line generation unit **2040** generates the queue line **40**, based on the position and the orientation being estimated for each object **20** included in the queue region **30** (S108).

(41) Note that, a flow of the processing performed by the queue analysis apparatus **2000** is not limited to the example illustrated in FIG. 4. For example, the object **20** being a target of estimation of a position and an orientation may not only be the object **20** included in the queue region **30**, but may also be all the object **20** included in the target image **10**. In this case, for example, the queue analysis apparatus **2000** may perform estimation of a position and an orientation of the object **20** in parallel with estimation of the queue region **30**, or may perform estimation of a position and an orientation of the object **20** before estimation of the queue region **30**.

(42) Further, as described above, the processing of determining the queue region **30** may be performed by an apparatus other than the queue analysis apparatus **2000**. In this case, the queue analysis apparatus **2000** acquires information representing the queue region **30** from another apparatus.

(43) <Acquisition of Target Image **10**: S102>

(44) The estimation unit **2020** acquires the target image **10** to be a processing target (S102). The target image **10** is any image acquired by capturing the object **20** by the camera **50**. For example, the camera **50** is a surveillance camera provided at a place where a crowd of the objects **20** needs to be surveyed.

(45) The camera **50** may be a video camera that generates a moving image, or a still camera that generates a still image. In the former case, the target image **10** is a video frame constituting a moving image generated by the camera **50**.

(46) Various methods of acquiring the target image **10** by the estimation unit **2020** can be used. For example, the estimation unit **2020** receives the target image **10** transmitted from the camera **50**. In addition, for example, the estimation unit **2020** accesses the camera **50**, and acquires the target image **10** stored in the camera **50**.

(47) Note that, the camera **50** may store the target image **10** in a storage apparatus provided outside the camera **50**. In this case, the estimation unit **2020** acquires the target image **10** by accessing the storage apparatus. Thus, in this case, the queue analysis apparatus **2000** and the camera **50** may not be communicably connected to each other.

(48) When a part or a whole of a function of the queue analysis apparatus **2000** is achieved by the camera **50**, the queue analysis apparatus **2000** acquires the target image **10** generated by the queue analysis apparatus **2000** itself. In this case, the target image **10** is stored in a storage apparatus (for example, the storage device **1080**) located inside the queue analysis apparatus **2000**, for example. Thus, the estimation unit **2020** acquires the target image **10** from the storage apparatus.

(49) A timing at which the estimation unit **2020** acquires the target image **10** is optional. For example, each time the target image **10** is generated by the camera **50**, the estimation unit **2020**

acquires the newly generated target image **10**. In addition, for example, the estimation unit **2020** may regularly acquire the target image **10** that has not yet been acquired. For example, when the estimation unit **2020** acquires the target image **10** once in a second, the estimation unit **2020** collectively acquires one or more target images **10** being generated in one second (for example, 30 target images **10** when the camera **50** is a video camera having a frame rate of 30 frames/second (fps)).

(50) The estimation unit **2020** may acquire all of the target images **10** generated by the camera **50**, or may acquire some of the target images **10**. In the latter case, for example, the estimation unit **2020** acquires the target image **10** generated by the camera **50** at a proportion of one to a predetermined number.

(51) <Estimation of Queue Region **30**>

(52) As described above, the estimation unit **2020** may perform processing of estimating the queue region **30** from the target image **10**. Herein, various existing techniques can be used as a technique for estimating a region representing a queue from an image.

(53) For example, an estimator that estimates a queue region included in an image is provided in the estimation unit **2020**, and the estimator is learned in advance. Note that, various kinds of models such as a neural network can be used as a model used as the estimator.

(54) The estimator is learned in such a way as to output information representing the queue region **30** included in the target image **10** in response to an input of the target image **10**. The information representing the queue region **30** is information that can determine a pixel included in the queue region **30** among pixels included in the target image **10**. For example, the information representing the queue region **30** is information listing coordinates of each pixel included in the queue region **30**, information indicating whether each pixel included in the target image **10** is included in the queue region **30**, and the like. In addition, for example, when the queue region **30** is represented by a polygon, the information representing the queue region **30** may indicate coordinates of each vertex of the polygon.

(55) An existing technique can be used for learning the estimator that estimates a queue region. For example, a plurality of pieces of learning data formed of a set of “a learning image, and information representing a queue region in the learning image” are prepared. Then, for each piece of the learning data, by performing processing of “inputting a learning image included in the learning data into an estimator, computing a loss representing an error between an output (a queue region estimated from the learning image by the estimator) of the estimator and the queue region indicated by the learning data, and updating a parameter of the estimator in such a way as to minimize the loss”, the estimator can be learned.

(56) As described above, the processing of estimating the queue region **30** from the target image **10** may be performed by an apparatus other than the queue analysis apparatus **2000**. In this case, the estimation unit **2020** acquires information representing the queue region **30** from the another apparatus. For example, the information is information in which information that determines each of one or more queue regions **30** estimated for the target image **10** is associated with identification information about the target image **10**.

(57) <Estimation of Position and Orientation of Object **20**>

(58) The estimation unit **2020** estimates a position and an orientation of the object **20** included in the target image **10**. Herein, an existing technique can be used as a technique for estimating, for an object such as a person included in an image, a position and an orientation of the object.

(59) For example, an estimator configured in such a way as to output a position and an orientation of each of one or more objects **20** included in the target image **10** in response to an input of the target image **10** is provided in the estimation unit **2020**. Then, the estimator is learned in advance by using learning data. In this case, the estimation unit **2020** can acquire a position and an orientation of each object **20** included in the target image **10** by inputting the target image **10** into the estimator. Note that, any model such as a neural network can be used as a model of the

estimator.

(60) In addition, for example, the estimation unit **2020** may estimate a position of the object **20** by detecting a predetermined feature value of the object **20** from the target image **10**. At this time, by preparing a feature value for each orientation of the object **20** as a feature value of the object **20**, not only estimation of a position of the object **20** but also estimation of an orientation of the object **20** can be performed.

(61) A position and an orientation of the object **20** may be estimated by a time-series analysis using not only one target image **10** but a plurality of the target images **10**. For example, a position and an orientation of the object **20** are represented by a state space model, and a parameter of the model is estimated by using the target images **10** in time series, and thus a model that can estimate a position and an orientation of the object **20** at a desired point in time can be acquired. Note that, various algorithms such as a Kalman filter and a particle filter can be used for estimation of a parameter.

(62) Herein, as described above, a group of a plurality of objects and a part of one object may be handled as the object **20**. In a case where a group of a plurality of objects are handled as one object **20**, for example, a technique using a crowd patch disclosed in Patent Document 2 can be used as a technique for estimating a position and an orientation of the object **20**.

(63) <Generation of Queue Line **40**>

(64) The queue line generation unit **2040** generates the queue line **40**. For example, generation of the queue line **40** can be achieved by a flow exemplified below.

(65) First, the queue line generation unit **2040** computes a direction vector representing a position and an orientation of each object **20**, based on a position and an orientation being estimated for each object **20** included in the queue region **30**. FIG. 5 is a diagram illustrating a direction vector computed for each object **20** included in the queue region **30**. For example, a direction vector **60** of a certain object **20** is computed as a vector in which a starting point of the direction vector **60** is a position estimated for the object **20**, and an orientation of the direction vector **60** is an orientation estimated for the object **20**.

(66) A length of a direction vector may be a fixed length predetermined for all direction vectors, or may be a length different for each direction vector. In the latter case, for example, a length of each direction vector **60** is determined in such a way that lengths of the direction vectors **60** are identical when the direction vectors **60** are mapped in the real world. In a case where the camera **50** faces obliquely below and performs capturing, lengths on the target image **10** vary depending on a position and a direction of objects even when the objects having the same length are captured. Thus, for example, a reference length of the direction vector **60** in the real world is predetermined. Then, the queue line generation unit **2040** computes the direction vector **60** associated with the object **20**, as a vector on the target image **10** to be a vector having a reference length drawn from a position of the direction vector **60** in an orientation of the direction vector **60** when the direction vector **60** is mapped in the real world, based on a position and an orientation being estimated for the object **20**.

(67) The queue line generation unit **2040** performs ordering of a direction vector. For example, three objects **20** are included in the queue region **30**, three direction vectors **60** are ordered by determining each of a front, a second, and a third from among the direction vectors **60**. Hereinafter, information representing ordering of the direction vector **60** is referred to as order information.

(68) Ordering of the direction vector **60** can be achieved by, for example, determining, for each direction vector **60**, another direction vector **60** adjacent to the direction vector **60** with reference to an end point of the direction vector **60**. Herein, another direction vector **A2** adjacent to a direction vector **A1** with reference to an end point is the direction vector **60** having a starting point thereof closest to an end point of the direction vector **A1** among the direction vectors **60** having at least a part thereof being included in the queue region **30**. The object **20** associated with the direction vector **A2** is the object **20** located right in front of the object **20** associated with the direction vector **A1** in a queue.

(69) In this way, by determining, for each direction vector **60**, another direction vector **60** adjacent to the direction vector **60** with reference to an end point of the direction vector **60**, the direction vector **60** located right in front (the direction vector **60** associated with the object **20** located right in front of the object **20** associated with the direction vector **60**) can be determined. Thus, each direction vector **60** can be ordered.

(70) FIG. **6** is a diagram illustrating order information in a table format. In FIG. **6**, a direction vector **60-1** is adjacent to a direction vector **60-2** with reference to an end point of the direction vector **60-1**. Further, the direction vector **60-2** is adjacent to a direction vector **60-3** with reference to an end point of the direction vector **60-2**. Thus, an order of the direction vectors **60** in a queue is the “direction vector **60-1**, the direction vector **60-2**, and the direction vector **60-3**” in an order from an end.

(71) The queue line generation unit **2040** generates the queue line **40** by using each direction vector **60** and the order information. For example, the queue line generation unit **2040** generates the queue line **40** by connecting the direction vectors **60** according to an order indicated in the order information. More specifically, the queue line generation unit **2040** connects, for each direction vector **60**, an end point of the direction vector **60** to a starting point of another direction vector **60** adjacent to the direction vector **60** with reference to the end point of the direction vector **60**.

(72) FIG. **7** is a diagram illustrating the queue line **40** generated by connecting the direction vectors **60**. Note that, in order to make the drawing clear, FIG. **7** illustrates the queue line **40** in a position slightly shifted from the direction vector **60** in such a way that the queue line **40** does not overlap the direction vector **60**. In an actual operation, the queue line **40** may overlap the direction vector **60**.

(73) Order information illustrated in FIG. **7** is the same as the order information illustrated in FIG. **6**. The order information indicates that the direction vector **60-1** is adjacent to the direction vector **60-2** with reference to the end point of the direction vector **60-1**. Then, the queue line generation unit **2040** connects the end point of the direction vector **60-1** to a starting point of the direction vector **60-2**. Further, the order information indicates that the direction vector **60-2** is adjacent to the direction vector **60-3** with reference to the end point of the direction vector **60-2**. Then, the queue line generation unit **2040** connects the end point of the direction vector **60-2** to a starting point of the direction vector **60-3**. As a result, the queue line **40** acquired by connecting the direction vector **60-1** to the direction vector **60-3** is generated.

(74) A method of generating the queue line **40** by using the direction vector **60** and the order information is not limited to the method described above. For example, the queue line generation unit **2040** may generate the queue line **40** by connecting predetermined positions (for example, a starting point, a middle point, an end point, or the like) of the direction vectors **60** adjacent to each other. FIG. **8** is a diagram illustrating the queue line **40** generated by connecting middle points of the direction vectors **60**. Order information is the same as that in the examples in FIGS. **6** and **7**. Thus, the queue line generation unit **2040** generates the queue line **40** by connecting a middle point of the direction vector **60-1** to a middle point of the direction vector **60-2**, and connecting a middle point of the direction vector **60-2** to a middle point of the direction vector **60-3**.

(75) The queue line generation unit **2040** may perform processing (so-called linear approximation) of generating one straight line passing through the vicinity of each point (such as a starting point, a middle point, and an end point) of the direction vector **60**, and set, as the queue line **40**, the one straight line acquired as a result of the processing. FIGS. **9A** and **9B** are diagrams illustrating the queue line **40** acquired by the linear approximation. In order to make the drawing clear, the object **20** is omitted from FIGS. **9A** and **9B**.

(76) FIG. **9A** illustrates a case where the linear approximation is applied in the example in FIG. **7**. The queue line **40** represents a queue line acquired by the linear approximation, and a zigzag dotted line represents the queue line **40** in FIG. **7**.

(77) FIG. **9B** illustrates a case where the linear approximation is applied in the example in FIG. **8**.

The queue line **40** represents a queue line acquired by the linear approximation, and a zigzag dotted line represents the queue line **40** in FIG. **8**.

(78) Note that, an existing technique can be used as a specific technique for computing one straight line passing through the vicinity of a plurality of points by the linear approximation.

(79) The queue line generation unit **2040** may perform non-linear interpolation in connection between the direction vectors **60**. FIGS. **10A** and **10B** are diagrams illustrating a case where the non-linear interpolation processing is performed in generation of the queue line **40**. FIG. **10A** illustrates a case where the non-linear interpolation processing is applied in the example in FIG. **7**. Specifically, in a case where the end point of the direction vector **60-1**, the starting point of the direction vector **60-2**, the end point of the direction vector **60-2**, and the starting point of the direction vector **60-3** are connected in this order, the queue line **40** is generated as a smooth curved line passing through points of the direction vectors or the vicinity thereof instead of being generated by simply connecting (linear-interpolating) the direction vectors.

(80) FIG. **10B** illustrates a case where the non-linear interpolation processing is applied in the example in FIG. **8**. Specifically, in a case where the middle point of the direction vector **60-1**, the middle point of the direction vector **60-2**, and the middle point of the direction vector **60-3** are connected, the queue line **40** is generated as a smooth curved line passing through points of the direction vectors or the vicinity thereof.

(81) Herein, various existing techniques can be used as a specific technique for computing a curved line passing through a plurality of specific points or the vicinity thereof by the non-linear interpolation.

(82) <<Shape that can be Used as Queue Line **40**>>

(83) As the queue line **40**, any linear shape can be adopted. FIGS. **11A** to **11E** are diagrams illustrating a shape that can be used as the queue line **40**. As the queue line **40**, any line such as a solid line (**40-1**), a dotted line (**40-2**), or a dot-and-dash line (**40-3**) can be adopted. Further, a hollow line (shape acquired by stretching out a rectangle) may be used as the queue line **40** (**40-4**). Further, the queue line **40** may be an arrow (**40-5**). Note that, a thickness of the queue line **40** is optional.

(84) <Output of Information>

(85) The queue analysis apparatus **2000** outputs various pieces of information related to the queue line **40**. Hereinafter, the information is referred to as output information. Further, a functional configuration unit that outputs the output information is referred to as an output unit. FIG. **12** is a block diagram illustrating a functional configuration of the queue analysis apparatus **2000** including an output information generation unit **2060**.

(86) The output information may include various pieces of information. Hereinafter, various pieces of information that may be included in the output information will be described.

(87) <<Provision of Queue Line **40**>>

(88) For example, the output information generation unit **2060** generates, as the output information, an image (hereinafter, a result image) in which the queue line **40** is superimposed on the target image **10**. FIG. **13** is a diagram illustrating a result image. In a result image **70** in FIG. **13**, the queue line **40** achieved as a solid line is superimposed on the target image **10**. Further, in this example, a queue region is also superimposed on the target image **10**. The queue region can be superimposed by any method in such a way that a boundary of a region is superimposed by a line, the inside of a region is superimposed by a semitransparent color, a region is emphasized by reducing a brightness value other than the region, or the like.

(89) Herein, the output information generation unit **2060** may superimpose the queue line **40** generated by the queue line generation unit **2040** only on the target image **10** used for generating the queue line **40**, or may further superimpose the queue line **40** on the target image **10** other than the target image **10** used for generating the queue line **40**. For example, the queue line generation unit **2040** is assumed to extract the target image **10** at a proportion of one in one second from a

video image at 30 frames/second being generated from the camera **50**, and generate the queue line **40** from the extracted target image **10**. When the queue line **40** is superimposed only on the target image **10** used for generation of the queue line **40**, the result image **70** is time-series data at one frame/second. In other words, the time-series result image **70** represents a change in a position of the object **20** and the queue line **40** by one frame/second.

(90) On the other hand, when the queue line **40** is also superimposed on the target image **10** other than the target image **10** used for generation of the queue line **40**, for example, the output information generation unit **2060** superimposes the queue line **40** generated from a certain target image **10** on not only the certain target image **10** but also on 29 target images **10** (each target image **10** before the target image **10** used for generation of the queue line **40** next) being generated after the certain target image **10**. In other words, one queue line **40** is superimposed on a video image of a second. In this case, the time-series result image **70** represents a change in a position of the object **20** by 30 frames/second, and represents a change in the queue line **40** by one frame/second.

(91) <<Provision of Front and End>>

(92) The output information may further include information representing one or both of a front position and an end position of a queue. FIG. **14** is a diagram illustrating the result image **70** indicating a front position and an end position of a queue. The result image **70** in FIG. **14** is acquired by superimposing, on the target image **10**, each of a front marker **72** representing a front position and an end marker **74** representing an end position.

(93) Various methods of determining a front position and an end position of a queue can be used. For example, the output information generation unit **2060** determines, as a front position of a queue, an end point associated with a front direction vector **60** in ordering of the direction vector **60** among end points of the queue line **40**. On the other hand, the output information generation unit **2060** determines, as an end position of a queue, an end point associated with an end direction vector **60** in ordering of the direction vector **60** among end points of the queue line **40**.

(94) Further, as in a queue formed in front of a checkout counter and the like, an approximate position of a front position of a queue may be clear in advance. In this case, a position predicted to be a front position of a queue is predetermined. Then, the output information generation unit **2060** determines, as a front position of the queue, an end point closer to the predicted position among end points of the queue line **40**, and determines another position as an end position of the queue.

(95) Herein, the result image **70** in FIGS. **13** and **14** includes a frame of a dotted line representing an outer frame of the queue region **30**. However, the result image **70** may not include a frame representing the queue region **30**.

(96) <<Provision of Waiting Time>>

(97) The output information generation unit **2060** may compute a waiting time in any position of a queue by using the queue line **40**. In this case, it is assumed that, in principle, the object **20** joins a queue from an end of the queue line **40**, advances from the end toward a front of the queue line **40**, and eventually leaves the queue from a front position. Then, a waiting time in a certain position of a queue refers to a time required since the object **20** reaches the certain position until the object **20** leaves from a front of the queue.

(98) For example, the output information generation unit **2060** computes a waiting time at an end of a queue. The waiting time at the end of the queue is computed based on a length of the queue line **40** or a size (area) of the queue region **30**, for example. In this case, a waiting time associated with a unit length of the queue line **40** or a unit area of the queue region **30** is predetermined. Then, the output information generation unit **2060** computes the waiting time at the end of the queue by multiplying the length of the queue line **40** by the waiting time associated with the unit length of the queue line **40** or by multiplying the area of the queue region **30** by the waiting time associated with the unit area of the queue region **30**.

(99) FIG. **15** is a diagram illustrating the result image **70** indicating a waiting time at an end. The result image **70** in FIG. **15** is acquired by further adding a waiting time display **90** representing a

waiting time at the end of the queue line **40** to the result image **70** in FIG. **14**.

(100) The queue analysis apparatus **2000** can compute a waiting time at an end of a queue even in a situation in which a place where a queue is to be formed is not predetermined and where and what kind of shape the queue is to be formed are not clear.

(101) A position (hereinafter, a target position) being a computation target of a waiting time is not limited to the end of the queue line **40**. When any position other than the end of the queue line **40** is handled as a target position, the output information generation unit **2060** computes a waiting time in the target position, based on a waiting time at an end of a queue. Specifically, first, the output information generation unit **2060** computes a length from a front of a queue to a target position along the queue line **40**. Then, the output information generation unit **2060** computes a ratio (the computed length/an entire length of the queue line **40**) between the computed length and the entire length of the queue line **40**. Then, the output information generation unit **2060** computes a waiting time in the target position by multiplying the waiting time at the end of the queue by the computed ratio. In addition, for example, a waiting time may be computed by multiplying a length of the queue line **40** from a front of a queue to a target position and a size (area) of the queue region **30** by a waiting time associated with a unit length of the queue line **40** or by a waiting time associated with a unit area of the queue region **30**.

(102) The queue analysis apparatus **2000** can compute a distance from a front of a queue to each position on the queue along the queue line **40**. Thus, as compared to a case where the queue line **40** is not used, a waiting time in any position of a queue can be more accurately recognized.

(103) A target position may be automatically determined by the output information generation unit **2060**, based on a predetermined reference, or may be specified by a user. In the former case, for example, the output information generation unit **2060** handles, as a target position, each division point when the queue line **40** is divided into a predetermined number, and an end. For example, when the predetermined number is three, two points are determined as division points at which the queue line **40** is divided into three. Thus, the output information generation unit **2060** handles, as a target point, each of the two points and an end, and computes a waiting time for the target point.

(104) When a target position is specified by a user, for example, the queue analysis apparatus **2000** displays a screen including the result image **70** on a display apparatus, and receives a user input for specifying a point on the queue line **40**. The output information generation unit **2060** handles, as a target position, a point specified in such a manner.

(105) FIG. **16** is a diagram illustrating the result image **70** also indicating a waiting time in a position other than an end. In FIG. **16**, the queue line **40** is divided into three by a division position **100** and a division position **110**. Then, in the result image **70**, a display **102** representing the division position **100**, a display **104** representing a waiting time in the division position **100**, a display **112** representing the division position **110**, and a display **114** representing a waiting time in the division position **110** are included.

(106) Note that, as described above, as a method of computing a waiting time, a waiting time is computed based on a length of the queue line **40** or a size (area) of the queue region **30**, but a waiting time may be similarly computed based on the number of people on the queue line **40**. For example, a waiting time in a target position can be computed by multiplying the number of people from a front to the target position by a waiting time (a time required since a person moves to the front of a queue until the person leaves the queue) per person. In this case, a waiting time per person is predetermined.

(107) <<Provision of Past Queue Line **40**>>

(108) The output information generation unit **2060** may include, in the result image **70**, one or more queue lines **40** computed in the past for the same queue as a queue represented by the queue line **40** (latest queue line **40**) generated by the queue line generation unit **2040** in addition to the queue line **40** generated by the queue line generation unit **2040**. Note that, the “same queue” herein refers to a queue formed at the same place, such as a queue present at points in time different from each other

in front of the same checkout counter, a queue in which the object **20** leaving from a front uses the same facility or the same place, and the like.

(109) FIG. **17** is a diagram illustrating the result image **70** including a past queue line **40**. FIG. **17** illustrates a display (such as “present” and “before X minutes”) representing a point in time associated with the queue line **40** by using lines of kinds different from each other for the queue lines **40** and near an end of the queue line **40** in such a way as to be able to identify each of the queue lines **40**. The point in time associated with the queue line **40** is, for example, a point in time at which the queue line **40** is generated, and a point in time at which the target image **10** used for generation of the queue line **40** is generated. The point in time associated with the queue line **40** may be represented by an absolute point in time (such as a date and time), or may be displayed by a relative point in time with reference to a predetermined point in time (for example, a present time). A case of FIG. **17** is the latter case.

(110) In this way, by also providing the past queue line **40** together, the present and past queue lines **40** can be easily compared, and thus various pieces of information (such as a change in length of a queue and a change in position of a queue) related to a queue can be easily acquired. For example, when it is clear that a length of a queue tends to increase, a need for action for relieving congestion can be recognized. In addition, for example, when it is clear that a position of a queue tends to move toward an unexpected position, a need for action for lining up a queue and the like can be recognized.

(111) Herein, various methods of deciding whether a plurality of the queue lines **40** at points in time different from each other represent the same queue can be used. For example, when the queue region **30** associated with the queue line **40** generated at a certain point in time and the queue region **30** associated with the queue line **40** generated at another point in time overlap each other, it is decided that the queues represented by the queue lines **40** are the same. At this time, when a threshold value is provided for a degree of overlapping of the queue region **30** and the degree of overlapping is equal to or more than the threshold value, it may be estimated that associated queue lines **40** represent the same queue.

(112) In addition, for example, when a distance between front positions of two queue lines **40** at points in time different from each other is equal to or less than a threshold value, the output information generation unit **2060** may decide that the queue lines **40** represent the same queue. According to the method, even when queues of the object **20** using the same facility or the same place are formed by extending in directions different from each other, the queues can be handled as the same queue.

(113) <<Provision of Queue Line **40** to Other than Target Image **10**>>

(114) In the description above, information about the queue line **40** and the like is superimposed on the target image **10**. However, information about the queue line **40** and the like may be superimposed on another image and the like.

(115) For example, the output information generation unit **2060** may generate the result image **70** by superimposing information about the queue line **40** and the like on an image acquired from the camera **50** at a time that does not include the object **20**. For example, in a case of a store and the like, by performing capturing by the camera **50** after business hours, an image that does not include the object **20** can be acquired. Such an image can be referred to as, for example, a background image.

(116) In addition, for example, the output information generation unit **2060** may generate the result image **70** by superimposing information about the queue line **40** and the like on a map image including a capturing range of the camera **50**. In this case, the queue line **40** needs to be mapped in an associated position on the map image. In other words, coordinates on the target image **10** need to be able to be converted into coordinates on the map image. Herein, an existing technique can be used as a technique for converting coordinates on an image generated by a camera into coordinates on a map image including a capturing range of the camera.

(117) <<Prediction of Future>>

(118) The output information generation unit **2060** may also generate, by using a plurality of the queue lines **40** generated for the same queue, a future queue line **40** for the queue. Specifically, the output information generation unit **2060** estimates a time change in the queue line **40** by using the plurality of queue lines **40** generated for the same queue, and generates the queue line **40** at a certain future point in time by using the estimation result.

(119) FIG. **18** is a diagram illustrating a method of estimating a future queue line **40**. In this example, the output information generation unit **2060** generates a queue line **40** (queue line **40-4**) after 15 minutes by using a present queue line **40** (queue line **40-1**), a queue line **40** (queue line **40-2**) before 15 minutes, and a queue line **40** (queue line **40-3**) before 30 minutes.

(120) Specifically, first, the output information generation unit **2060** divides each of the queue line **40-1** to the queue line **40-3** into four. Herein, in each of the queue lines **40**, positions at which the queue line **40** is divided are referred to as a second division position, a third division position, and a fourth division position in an order in which the positions are closer to a front.

(121) The output information generation unit **2060** estimates a time change in each of the division positions and an end position by using the present and past queue lines **40**. For example, the output information generation unit **2060** estimates a time change in the second division position (for example, generates a function representing the time change), based on the second division position of the queue line **40-3**, the second division position of the queue line **40-2**, and the second division position of the queue line **40-1**. Then, the output information generation unit **2060** computes (estimates) the second division position at a point in time desired to be predicted (after 15 minutes in a case of FIG. **18**), based on the point in time desired to be predicted and the estimated time change in the second division position (for example, by inputting the point in time desired to be predicted into the generated function).

(122) Similarly, the output information generation unit **2060** also estimates, for each of the third division position, the fourth division position, and the end position, a time change thereof, and computes (estimates) the third division position, the fourth division position, and the end position at a point in time desired to be predicted, based on the point in time desired to be predicted and the estimated time change.

(123) Then, the output information generation unit **2060** generates the queue line **40-4** at the point in time desired to be predicted by connecting the front position, the estimated second division position, the estimated third division position, the estimated fourth division position, and the estimated end position. Note that, as a method of generating the queue line **40** by connecting a plurality of positions, as described above, various methods such as a method of acquiring the queue line **40** by simple connection, a method of acquiring the queue line **40** by linear approximation, and a method of acquiring the queue line **40** by non-linear interpolation can be adopted.

(124) Note that, it is assumed that the front position does not move in the example in FIG. **18**, but the front position may also move similarly to the end position and the like. In this case, for example, the output information generation unit **2060** estimates a time change for the front position similarly to the end position and the like, and estimates a future front position, based on the estimated time change.

(125) Herein, a future point in time at which the queue line **40** is predicted may be a point in time (for example, after 15 minutes, after an hour, and the like) having a predetermined relationship with a present time, or may be any point in time specified by a user. In the latter case, the queue analysis apparatus **2000** receives an input indicating a point in time desired to be predicted from a user, and generates the queue line **40** for the specified point in time.

(126) A method of generating the queue line **40** for the future is not limited to the method described above. For example, an estimator learned in such a way as to output a future queue line **40** in response to an input of present and past queue lines **40** may be provided. For example, it is assumed that an estimator that generates the queue line **40** at a predicted point in time in response

to an input of a “present queue line **40**, a queue line **40** before 15 minutes, a queue line **40** before 30 minutes, and the predicted point in time” is provided. In this case, the estimator is learned by using, as learning data, a set of the “present queue line **40**, the queue line **40** before 15 minutes, the queue line **40** before 30 minutes, the predicted point in time, and the queue line **40** at the predicted point in time”. Specifically, the “present queue line **40**, the queue line **40** before 15 minutes, the queue line **40** before 30 minutes, and the predicted point in time” is input into the estimator, a loss representing an error between the queue line **40** output from the estimator and the queue line **40** at the predicted point in time being included in the learning data, and a parameter of the estimator is updated based on the computed loss.

(127) While the example embodiments of the present invention have been described with reference to the drawings, the example embodiments are only exemplification of the present invention, and various configurations other than the above-described example embodiments can also be employed.

(128) A part or the whole of the above-described example embodiments may also be described as in supplementary notes below, which is not limited thereto. 1. A queue analysis apparatus, including: an estimation unit that estimates a position and an orientation of an object included in a target image; and a queue line generation unit that generates, for a queue region being a region including a queue of an object in the target image, a queue line representing a queue included in the queue region by a line, based on the position and the orientation being estimated for the object included in the queue region. 2. The queue analysis apparatus according to supplementary note 1, wherein the queue line generation unit performs, on a direction vector representing a position and an orientation of each object constituting the queue, ordering based on an adjacent relationship between direction vectors, and generates the queue line, based on the direction vector of the each object constituting the queue and an order of the direction vector. 3. The queue analysis apparatus according to supplementary note 2, wherein the queue line generation unit generates, for each direction vector, a line connecting an end point of the direction vector and a starting point of another direction vector located right in front of the direction vector, and generates the queue line by the generated each line and the each direction vector. 4. The queue analysis apparatus according to supplementary note 2, wherein the queue line generation unit generates the queue line by performing linear approximation or non-linear interpolation on the ordered direction vector. 5. The queue analysis apparatus according to any one of supplementary notes 1 to 4, further including an output information generation unit that generates output information including a result image in which the queue line is included. 6. The queue analysis apparatus according to supplementary note 5, wherein the result image is acquired by superimposing the queue line on any one of the target image, a background image in which a same range as that of the target image is included, and a map image in which a same range as that of the target image is included. 7. The queue analysis apparatus according to supplementary note 5 or 6, wherein the output information generation unit estimates, for a queue represented by the queue line, any one or more of a front and an end of the queue, based on a position and an orientation of an object in a queue region in which the queue is included, and includes, in the result image, a display representing any one or more of the estimated front and the estimated end. 8. The queue analysis apparatus according to any one of supplementary notes 5 to 7, wherein the output information generation unit computes a distance along the queue line between a front of the queue line and another position on the queue line, computes a waiting time of the queue in the another position, based on the computed distance, and includes the computed waiting time in the output information. 9. The queue analysis apparatus according to any one of supplementary notes 5 to 8, wherein the output information generation unit includes, in the result image, a queue line generated in the past for a queue represented by the queue line generated from the target image. 10. The queue analysis apparatus according to any one of supplementary notes 6 to 9, wherein the output information generation unit estimates a time change in a queue line, based on the queue line generated from the target image and a queue line generated in the past for a queue represented by the queue line, generates a future queue line for the queue, based on the estimated

time change in the queue line, and includes the generated future queue line in the result image. 11. A control method executed by a computer, the control method including: an estimation step of estimating a position and an orientation of an object included in a target image; and a queue line generation step of generating, for a queue region being a region including a queue of an object in the target image, a queue line representing a queue included in the queue region by a line, based on the position and the orientation being estimated for the object included in the queue region. 12 The control method according to supplementary note 11, further including, in the queue line generation step, performing, on a direction vector representing a position and an orientation of each object constituting the queue, ordering based on an adjacent relationship between direction vectors, and generating the queue line, based on the direction vector of the each object constituting the queue and an order of the direction vector. 13. The control method according to supplementary note 12, further including, in the queue line generation step, generating, for each direction vector, a line connecting an end point of the direction vector and a starting point of another direction vector located right in front of the direction vector, and generating the queue line by the generated each line and the each direction vector. 14. The control method according to supplementary note 12, further including, in the queue line generation step, generating the queue line by performing linear approximation or non-linear interpolation on the ordered direction vector. 15. The control method according to any one of supplementary notes 11 to 14, further including an output information generation step of generating output information including a result image in which the queue line is included. 16 The control method according to supplementary note 15, wherein the result image is acquired by superimposing the queue line on any one of the target image, a background image in which a same range as that of the target image is included, and a map image in which a same range as that of the target image is included. 17. The control method according to supplementary note 15 or 16, further including, in the output information generation step, estimating, for a queue represented by the queue line, any one or more of a front and an end of the queue, based on a position and an orientation of an object in a queue region in which the queue is included, and including, in the result image, a display representing any one or more of the estimated front and the estimated end. 18. The control method according to any one of supplementary notes 15 to 17, further including, in the output information generation step, computing a distance along the queue line between a front of the queue line and another position on the queue line, computing a waiting time of the queue in the another position, based on the computed distance, and including the computed waiting time in the output information. 19. The control method according to any one of supplementary notes 15 to 18, further including, in the output information generation step, including, in the result image, a queue line generated in the past for a queue represented by the queue line generated from the target image. 20. The control method according to any one of supplementary notes 16 to 19, further including, in the output information generation step, estimating a time change in a queue line, based on the queue line generated from the target image and a queue line generated in the past for a queue represented by the queue line, generating a future queue line for the queue, based on the estimated time change in the queue line, and including the generated future queue line in the result image. 21. A program causing a computer to execute the control method according to any one of supplementary notes 11 to 20.

REFERENCE SIGNS LIST

(129) **10** Target image **20** Object **30** Queue region **40** Queue line **50** Camera **60** Direction vector **70** Result image **72** Front marker **74** End marker **90** Waiting time display **100** Division position **102** Display **104** Display **110** Division position **112** Display **114** Display **1000** Computer **1020** Bus **1040** Processor **1060** Memory **1080** Storage device **1100** Input/output interface **1120** Network interface **2000** Queue analysis apparatus **2020** Estimation unit **2040** Queue line generation unit **2060** Output information generation unit

Claims

1. A queue analysis apparatus, comprising: at least one memory configured to store instructions; and at least one processor configured to execute the instructions to perform operations comprising: estimating a position and an orientation of an object included in a target image; generating, for a queue region being a region including a queue of an object in the target image, a queue line representing a queue included in the queue region by a line, based on the position and the orientation being estimated for the object included in the queue region; and generating output information including a result image in which the queue line is included, wherein the result image is acquired by superimposing the queue line on any one of the target image, a background image in which a same range as that of the target image is included, and a map image in which a same range as that of the target image is included, and the operations further comprise: estimating a time change in a queue line, based on the queue line generated from the target image and a queue line generated in the past for a queue represented by the queue line; generating a future queue line for the queue, based on the estimated time change in the queue line; and including the generated future queue line in the result image.
2. The queue analysis apparatus according to claim 1, wherein the operations comprise: performing, on a direction vector representing a position and an orientation of each object constituting the queue, ordering based on an adjacent relationship between direction vectors; and generating the queue line, based on the direction vector of the each object constituting the queue and an order of the direction vector.
3. The queue analysis apparatus according to claim 2, wherein the operations comprise: generating, for each direction vector, a line connecting an end point of the direction vector and a starting point of another direction vector located right in front of the direction vector; and generating the queue line by the generated each line and the each direction vector.
4. The queue analysis apparatus according to claim 2, wherein the operations comprise generating the queue line by performing linear approximation or non-linear interpolation on the ordered direction vector.
5. The queue analysis apparatus according to claim 1, wherein the operations comprise: estimating, for a queue represented by the queue line, any one or more of a front and an end of the queue, based on a position and an orientation of an object in a queue region in which the queue is included; and including, in the result image, a display representing any one or more of the estimated front and the estimated end.
6. The queue analysis apparatus according to claim 1, wherein the operations comprise: computing a distance along the queue line between a front of the queue line and another position on the queue line; computing a waiting time of the queue in the another position, based on the computed distance; and including the computed waiting time in the output information.
7. The queue analysis apparatus according to claim 1, wherein the operations comprise including, in the result image, a queue line generated in the past for a queue represented by the queue line generated from the target image.
8. A control method executed by a computer, the control method comprising: estimating a position and an orientation of an object included in a target image; generating, for a queue region being a region including a queue of an object in the target image, a queue line representing a queue included in the queue region by a line, based on the position and the orientation being estimated for the object included in the queue region; and generating output information including a result image in which the queue line is included, wherein the result image is acquired by superimposing the queue line on any one of the target image, a background image in which a same range as that of the target image is included, and a map image in which a same range as that of the target image is included, and the control method further comprises: estimating a time change in a queue line, based

on the queue line generated from the target image and a queue line generated in the past for a queue represented by the queue line, generating a future queue line for the queue, based on the estimated time change in the queue line; and including the generated future queue line in the result image.

9. The control method according to claim 8, further comprising, performing, on a direction vector representing a position and an orientation of each object constituting the queue, ordering based on an adjacent relationship between direction vectors, and generating the queue line, based on the direction vector of the each object constituting the queue and an order of the direction vector.

10. The control method according to claim 9, further comprising, generating the queue line by performing linear approximation or non-linear interpolation on the ordered direction vector.

11. The control method according to claim 8, further comprising, estimating, for a queue represented by the queue line, any one or more of a front and an end of the queue, based on a position and an orientation of an object in a queue region in which the queue is included, and including, in the result image, a display representing any one or more of the estimated front and the estimated end.

12. The control method according to claim 8, further comprising, computing a distance along the queue line between a front of the queue line and another position on the queue line, computing a waiting time of the queue in the another position, based on the computed distance, and including the computed waiting time in the output information.

13. The control method according to claim 8, further comprising, including, in the result image, a queue line generated in the past for a queue represented by the queue line generated from the target image.

14. A non-transitory computer-readable medium storing a program causing a computer to execute a control method, the control method comprising: estimating a position and an orientation of an object included in a target image; generating, for a queue region being a region including a queue of an object in the target image, a queue line representing a queue included in the queue region by a line, based on the position and the orientation being estimated for the object included in the queue region; and generating output information including a result image in which the queue line is included, wherein the result image is acquired by superimposing the queue line on any one of the target image, a background image in which a same range as that of the target image is included, and a map image in which a same range as that of the target image is included, and the operations further comprise: estimating a time change in a queue line, based on the queue line generated from the target image and a queue line generated in the past for a queue represented by the queue line; generating a future queue line for the queue, based on the estimated time change in the queue line; and including the generated future queue line in the result image.
